

QUESTION No #1

i) $y_p = Ax^2 + Bx + C$

$y_L = (Ex + f)e^x$

$y_p = Ax^2 + Bx + C + (Ex + f)e^x$

$m = 4 \pm \sqrt{15}$

$y_c = c_1 e^{4+\sqrt{15}} + c_2 e^{4-\sqrt{15}}$

ii) $m^2 + 4 = 0$
 $m = \pm 2i$

$y_c = c_1 \cos 2x + c_2 \sin 2x$

$y_p = (Ax^2 + Bx + C) \sin 2x + (Dx^2 + Ex + f) \cos 2x$

$y_p = (Ax^3 + Bx^2 + Cx) \sin 2x + (Dx^3 + Ex^2 + fx) \cos 2x$

iii) $m^2 - 2m + 5 = 0$
 $D^2 - 2D + 5 = 0$

$D = 1 \pm 2i$

$y_c = c_1 e^x \cos 2x + c_2 e^x \sin 2x$
 $= (c_1 \cos 2x + c_2 \sin 2x) e^x$

$y_p = (A \cos 2x + B \sin 2x) e^x$

$y_p = (Ax \cos 2x + Bx \sin 2x) e^x$

Repeating Root

v) $D^2 - 2D + 2 = 0$

$D = 1 \pm i$

$y_c = c_1 e^x \cos x + c_2 e^x \sin x$
 $= (c_1 \cos x + c_2 \sin x) e^x$

$y_p = e^{2x} (A \cos 2x + B \sin 2x + C \sin x + D \cos x)$

$y_p = Ae^{2x} \cos 2x + Be^{2x} \sin 2x + Ce^{2x} \sin x + De^{2x} \cos x$

$$v) \quad D^3 - D^2 - 4D + 4 = 0$$

$$D = 1, 2, -2 \quad (\text{calculator})$$

$$y_c = c_1 e^x + c_2 e^{2x} + c_3 e^{-2x}$$

$$y_p = A + B e^x + C e^{2x}$$

— Repeating

$$y_p = A + B x e^x + C x e^{2x}$$

~~same~~ root

$$vi) \quad D^2 - 4D + 8 = 0$$

$$D = 2 \pm 2i$$

$$y_c = c_1 e^{2x} \cos 2x + c_2 e^{2x} \sin 2x$$

$$= (c_1 \cos 2x + c_2 \sin 2x) e^{2x}$$

$$y_p = (Ax^2 + Bx + C) e^{2x} \sin 2x + (Dx^2 + Ex + f) e^{2x} \cos 2x$$

— Repeating (root)

$$= (Ax^3 + Bx^2 + Cx) e^{2x} \sin 2x + (Dx^3 + Ex^2 + f x) e^{2x} \cos 2x$$

QUESTION No # 2

$$y'' + 6y' + 9y = 6x^2 + 2 - 12e^{3x} - \sin 2x$$

$$D^2 + 6D + 9 = 0$$

$$D = -3, -3$$

$$y_c = c_1 e^{-3x} + c_2 x e^{-3x}$$

$$y_{p1} = Ax^2 + Bx + C$$

$$y_{p1}' = 2Ax + B$$

$$y_{p1}'' = 2A$$

$$y_{p2} = D e^{3x}$$

$$y_{p2}' = 3D e^{3x}$$

$$y_{p2}'' = 9D e^{3x}$$

$$y_{p3} = E \sin 2x + f \cos 2x$$

$$y_{p3}' = 2E \cos 2x - 2f \sin 2x$$

$$y_{p3}'' = -4E \sin 2x - 4f \cos 2x$$

Put in eq. and comparing,

$$2A + 6(2Ax + B) + 9(Ax^2 + Bx + C) = 6x^2 + 2$$

$$2A + 12Ax + 6B + 9Ax^2 + 9Bx + 9C = 6x^2 + 2$$

$$9A = 6 \quad ; \quad A = 6/9 \quad ; \quad A = 2/3$$

$$U = \frac{x^2 \cdot 1}{x} + \frac{-x^1(1)}{x} + \frac{2}{x}$$

QUESTION No 3

a) $x^2 y'' - xy' + y = \ln x$
looks like cauchy euler,

$\gamma(x) = \frac{\ln x}{x^2}$

so, let $y = x^m$

$y' = m x^{m-1}$

$y'' = m(m-1) x^{m-2}$

$x^2 \cdot x^{m-2} (m^2 - m) - x \cdot x^{m-1} m + x^m = 0$

$x^m (m^2 - m) - x^m m + x^m = 0$

$x^m (m^2 - m - m + 1) = 0$

$m^2 - 2m + 1 = 0$

$m = 1, 1$

$y_c = c_1 x + c_2 x \ln x$

$y_1 = x, \quad y_2 = x \ln x$

$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

$= \begin{vmatrix} x & x \ln x \\ 1 & \ln x + 1 \end{vmatrix}$

$= x(\ln x + 1) - x \ln x$
 $= x \cancel{\ln x} + x - x \cancel{\ln x}$

$\boxed{W = x}$

$u = - \int \frac{y_2 \gamma(x)}{W}$

$= - \int \frac{x \ln x \frac{\ln x}{x^2}}{x}$

$= - \int \frac{(\ln x)^2}{x^2}$

$= \frac{(\ln x)^2}{x} - \int -\frac{1}{x} (-2 \ln(x)) \frac{1}{x}$

$= \frac{(\ln x)^2}{x} - \int \frac{1}{x^2} (2 \ln(x))$

$= \frac{(\ln x)^2}{x} - \left[2 \ln(x) \cdot \frac{1}{x^2} - \int -\frac{1}{x} \left(\frac{2}{x} \right) \right]$

$u = \frac{(\ln x)^2}{x} + \frac{2 \ln(x)}{x} + \frac{2}{x}$

$$12A + 9B = 0$$

$$12\left(\frac{2}{3}\right) + 9B = 0$$

$$8 + 9B = 0$$

$$\boxed{B = -\frac{8}{9}}$$

$$y_{p1} = \frac{2}{3}x^2 - \frac{8}{9}x + \frac{2}{3}$$

Now, comparing y_{p2}

$$9De^{3x} + 6(3De^{3x}) + 9(De^{3x}) = -12e^{3x} + 2$$

$$9De^{3x} + 18De^{3x} + 9De^{3x} = -12e^{3x} + 2$$

$$36De^{3x} = -12e^{3x}$$

$$36D = -12$$

$$D = \frac{-12}{36} = -\frac{1}{3}$$

$$\boxed{y_{p2} = -\frac{1}{3}e^{3x}}$$

Now, comparing y_{p3}

$$-4E\sin 2x - 4f\cos 2x + 6(2E\cos 2x - 2f\sin 2x) + 9(E\sin 2x + f\cos 2x) = -\sin 2x$$

$$-4E\sin 2x - 4f\cos 2x + 12E\cos 2x - 12f\sin 2x + 9E\sin 2x + 9f\cos 2x = -\sin 2x$$

$$-4E - 12f + 9E = -1$$

$$5E - 12f = -1$$

$$\boxed{E = -\frac{5}{169}}$$

$$-4f + 12E + 9f = 0$$

$$5f + 12E = 0$$

$$\boxed{f = \frac{12}{169}}$$

$$y_{p3} = \frac{-5}{169}\sin 2x + \frac{12}{169}\cos 2x$$

$$= y_c + y_p$$

$$y = c_1e^{-3x} + c_2xe^{3x} + \frac{2}{3}x^2 - \frac{8}{9}x + \frac{2}{3} - \frac{1}{3}e^{3x} - \frac{5}{169}\sin 2x + \frac{12}{169}\cos 2x$$

$$2A + 6B + 9C = 2$$

$$2\left(\frac{2}{3}\right) + 6\left(-\frac{8}{9}\right) + 9C = 2$$

$$\frac{4}{3} - \frac{16}{3} + 9C = 2$$

$$-\frac{12}{3} + 9C = 2$$

$$9C = 2 + \frac{12}{3}$$

$$9C = 6$$

$$\boxed{C = \frac{2}{3}}$$

$$v = \int \frac{y_1 r(x)}{w}$$

$$= \int \frac{x \frac{\ln x}{x^2}}{x}$$

$$= \int \frac{\ln x}{x^2}$$

$$v = -\frac{\ln x}{x} - \frac{1}{x}$$

$$y_p = y_1 u + y_2 v$$

$$= x \left(\frac{(\ln x)^2}{x} + \frac{2 \ln x}{x} + \frac{2}{x} \right) + x \ln x \left(-\frac{\ln x}{x} - \frac{1}{x} \right)$$

$$= (\ln x)^2 + 2 \ln x + 2 - (\ln x)^2 - \ln x$$

$$y_p = \ln x + 2$$

$$b) x^2 y'' + x y' + y = \sec(\ln x)$$

$$y = x^m$$

$$y' = m x^{m-1}$$

$$y'' = (m^2 - m) x^{m-2}$$

$$x^m (m^2 - m + 1) = 0$$

$$x^m \neq 0$$

$$m^2 + 1 = 0$$

$$m = \pm i$$

$$y_c = c_1 \cos(\ln x) + c_2 \sin(\ln x)$$

$$y_1 = \cos(\ln x), y_2 = \sin(\ln x)$$

$$r(x) = \frac{\sec(\ln x)}{x^2}$$

$$w = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$= \begin{vmatrix} \cos(\ln x) & \sin(\ln x) \\ -\frac{1}{x} \sin(\ln x) & \frac{1}{x} \cos(\ln x) \end{vmatrix}$$

$$= \frac{\cos^2(\ln x)}{x} + \frac{\sin^2(\ln x)}{x}$$

$$= \frac{\cos^2(\ln x) + \sin^2(\ln x)}{x}$$

$$w = \frac{1}{x}$$

$$u = -\int \frac{y_2 r(x)}{w} = -\int \frac{\sin(\ln x) \frac{\sec(\ln x)}{x^2}}{\frac{1}{x}} = \int \frac{-\tan(\ln x)}{x} = \ln(\cos(\ln x))$$

$$v = \int \frac{y_1 r(x)}{w} = \int \frac{\cos(\ln x) \frac{\sec(\ln x)}{x^2}}{\frac{1}{x}} = \int \frac{1}{x} = \ln x$$

$$y_p = y_1 u + y_2 v$$

$$= \cos(\ln x) (\ln(\cos(\ln x))) + \ln x (\sin(\ln x))$$

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x) + \cos(\ln x) (\ln(\cos(\ln x))) + \ln x (\sin(\ln x))$$

$$x^3 y''' - 3x^2 y'' + 6xy' - 6y = 3 + \ln x^3$$

$$y = x^m$$

$$y' = mx^{m-1}$$

$$y'' = m(m-1)x^{m-2} = (m^2 - m)x^{m-2}$$

$$y''' = m(m-1)(m-2)x^{m-3} = (m^2 - m)(m-2)x^{m-3}$$

$$x^m (m^3 - 3m^2 - 2m) - 3x^m (m^2 - m) + 6x^m (m) - 6x^m = 0$$

$$x^m (m^3 - 3m^2 - 2m - 3m^2 + 3m + 6m - 6) = 0$$

$$x^m (m^3 - 6m^2 + 7m - 6) = 0$$

$$x^m \neq 0$$

$$m^3 - 6m^2 + 7m - 6 = 0$$

$$m = 1, 2, 3$$

$$y_c = c_1 x + c_2 x^2 + c_3 x^3$$

$$y_1 = x, y_2 = x^2, y_3 = x^3$$

$$x(x) = \frac{3 + 3 \ln x}{x^3}$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ x(x) & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & x^2 & x^3 \\ 0 & 2x & 3x^2 \\ x(x) & 2 & 6x \end{vmatrix}$$

$$= x(x) (3x^4 - 2x^4) = x(x) (x^4) = \frac{3 + 3 \ln x}{x^3} x^4$$

$$W_1 = 3x + 3x \ln x$$



$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & r(u) & y_3'' \end{vmatrix} = \begin{vmatrix} x & 0 & x^3 \\ 1 & 0 & 3x^2 \\ 0 & r(u) & 6x \end{vmatrix}$$

$$= -r(u)(3x^3 - x^3)$$

$$= -r(u)(2x^3)$$

$$= \textcircled{3x} - \left(\frac{3 + 3 \ln x}{x^3} \right) (2x^3)$$

$$W_2 = -6 - 6 \ln x$$

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & r(u) \end{vmatrix}$$

$$= \begin{vmatrix} x & x^2 & 0 \\ 1 & 2x & 0 \\ 0 & 2 & r(u) \end{vmatrix}$$

$$= r(x)(2x^2 - x^2)$$

$$= \frac{3 + 3 \ln x}{x^3} (x^2)$$

$$W_3 = \frac{3x + 3x \ln x}{x}$$

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$$

$$= x(12x^2 - 6x^2) - (6x^3 - 2x^3)$$

$$= x(6x^2) - 4x^3$$

$$= 6x^3 - 4x^3$$

$$W = 2x^3$$

$$U_1 = \int \frac{W_1}{W} = \int \frac{3x + 3x \ln x}{2x^3} = \int \frac{3x}{2x^3} + \int \frac{3x \ln x}{2x^3}$$

$$= \int \frac{3}{2x^2} + \int \frac{3 \ln x}{2x^2} = -\frac{3}{2x} + \frac{3}{2} \left[-\frac{\ln x}{x} - \int -\frac{1}{x^2} \right]$$

$$= -\frac{3}{2x} + \frac{3}{2} \left(-\frac{\ln x}{x} - \frac{1}{x} \right) = -\frac{3}{2x} - \frac{3 \ln x}{2x} - \frac{3}{2x}$$

$$U_1 = -\frac{3}{x} - \frac{3 \ln x}{x}$$

$$U_2 = \int \frac{W_2}{W} = \int \frac{-6 - 6 \ln x}{2x^3} = \int \frac{-6}{2x^3} - \frac{6}{2} \int \frac{\ln x}{x^3} = \frac{3}{2x^2} - 3 \left[\frac{\ln x}{-2x^2} - \int \frac{1}{-2x^3} \right]$$

$$U_2 = \frac{3}{2x^2} + \frac{3 \ln x}{2x^2} + \frac{3}{4x^2}$$

$$u_3 = \int \frac{w_3}{w_1} = \int \frac{3 + 3 \ln x}{2x^4} = \int \frac{3}{2x^4} + \int \frac{3 \ln x}{2x^4}$$

$$u_3 = \frac{-1}{2x^3} + \frac{3}{2} \left(-\frac{\ln x}{3x^3} - \int \frac{-1}{3x^4} \right)$$

$$u_3 = \frac{-1}{2x^3} + \frac{3}{2} \left(\frac{\ln x}{3x^3} - \frac{1}{9x^3} \right)$$

$$u_3 = \frac{-1}{2x^3} - \frac{\ln(x)}{2x^3} - \frac{1}{6x^3}$$

$$y_p = u_1 y_1 + u_2 y_2 + u_3 y_3$$

$$y_p = x \left(-\frac{3}{x} - \frac{3 \ln x}{2x} \right) + x^2 \left(\frac{3}{2x^2} + \frac{3 \ln x}{2x^2} - \frac{3}{4x^2} \right) + x^3 \left(\frac{-1}{2x^3} - \frac{\ln x}{2x^3} - \frac{1}{6x^3} \right)$$

$$y_p = -3 - \frac{3 \ln x}{2} + \frac{3}{2} + \frac{3 \ln x}{2} - \frac{3}{4} - \frac{1}{2} - \frac{\ln x}{2} - \frac{1}{6}$$

$$y_p = -\frac{17}{12} - \frac{\ln x}{2}$$

$$y = y_c + y_p$$

$$y = c_1 x + c_2 x^2 + c_3 x^3 - \frac{17}{12} - \frac{\ln x}{2}$$

QUESTION NO # 4

$$a) y'' - 2y' + y = \frac{e^t}{t^2}$$

$$D^2 - 2D + 1 = 0$$

$$D = 1, 1$$

$$y_c = c_1 e^t + c_2 t e^t$$

$$y_1 = e^t, \quad y_2 = t e^t$$

~~W~~

$$U = - \int \frac{y_2 r(n)}{w}$$

$$= - \int \frac{t e^t}{e^{2t}} \cdot \frac{e^t}{t^2} = - \ln t$$

~~W~~ ~~U~~

$$V = \int \frac{y_1 r(n)}{w} = \int \frac{e^t}{e^{2t}} \cdot \frac{e^t}{t^2} = - \frac{1}{t}$$

$$y = c_1 e^t + c_2 t e^t - \ln t (e^t) - \frac{1}{t} (t e^t)$$

$$y = c_1 e^t + c_2 t e^t - e^t \ln t - e^t$$

$$y_c = c_1 e^t + c_2 t e^t$$

$$y_c' = c_1 e^t + c_2 (e^t + t e^t)$$

Is it a IVP...

$$y = -e^t \ln t - e^{-t}$$

$$w = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^t & t e^t \\ e^t & e^t + t e^t \end{vmatrix} = e^{2t} + t e^{2t} - t e^{2t} = e^{2t}$$

$$w = e^{2t}$$

→ $\ln(0)$
not solvable
with given
IVP. . . .

$$b) \quad y'' + y = 3 \sin^2 x$$

$$D^2 + 1 = 0$$

$$D = \pm i$$

~~$$y_c = C_1 \sin x + C_2 \cos x$$~~

$$y_c = C_1 \sin x + C_2 \cos x$$

$$y_1 = \sin x, \quad y_2 = \cos x$$

$$U = - \int \frac{y_2 r(x)}{W} = - \int \frac{\cos x \cdot 3 \sin^2 x}{-1}$$

$$= \frac{3}{2} \int \sin 2x \sin x$$

$$= \frac{3}{2} \int \frac{1}{2} (\cos 3x - \cos x)$$

$$= -\frac{3}{4} \int \cos 3x - \cos x$$

$$U = -\frac{1}{4} \sin 3x + \frac{3}{4} \sin x$$

$$W = \begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix}$$

$$= -\sin^2 x - \cos^2 x$$

$$= -(\sin^2 x + \cos^2 x)$$

$$W = -1$$

$$= 3 \int \cos x \sin^2 x$$

$$V = \int \frac{y_1 r(x)}{W} = \int \frac{\sin x \cdot 3 \sin^2 x}{-1}$$

$$= -3 \int \sin x \left(\frac{1 - \cos 2x}{2} \right)$$

$$= -\frac{3}{2} \int \sin x - \sin x \cos 2x$$

$$= \frac{3}{2} \cos x + \frac{3}{2} \int \sin 3x - \sin x$$

$$= \frac{3}{2} \cos x + \frac{3}{2} \cos x - \frac{1}{2} \cos 3x$$

$$V = 3 \cos x - \frac{\cos 3x}{2}$$

~~$$y_p = \frac{1}{4} \sin -\frac{1}{4} \sin^2 3x + \frac{3}{4} \sin^2 x + 3 \cos^2 x - \frac{\cos^2 3x}{2}$$~~

~~$$y = C_1 \sin x + C_2 \cos x - \frac{1}{4} \sin^2 3x + \frac{3}{4} \sin^2 x + 3 \cos^2 x - \frac{\cos^2 3x}{2}$$~~

$$c) y''' - 3y'' + 2y' = \frac{e^{2x}}{1+e^x}$$

$$D^3 - 3D^2 + 2D = 0$$

$$D = 0, 1, 2$$

$$y_c = c_1 + c_2 e^x + c_3 e^{2x}$$

$$y_1 = e^0 = 1, y_2 = e^x, y_3 = e^{2x}$$

$$W = \begin{vmatrix} 1 & e^x & e^{2x} \\ 0 & e^x & 2e^{2x} \\ 0 & e^x & 4e^{2x} \end{vmatrix}$$

$$= 4e^{3x} - 2e^{3x}$$

$$W = 2e^{3x}$$

$$W_1 = \begin{vmatrix} 0 & e^x & e^{2x} \\ 0 & e^x & 2e^{2x} \\ x(x) & e^x & 4e^{2x} \end{vmatrix}$$

$$W_2 = \begin{vmatrix} 1 & 0 & e^{2x} \\ 0 & 0 & 2e^{2x} \\ 0 & x(x) & 4e^{2x} \end{vmatrix}$$

$$W_3 = \begin{vmatrix} 1 & e^x & 0 \\ 0 & e^x & 0 \\ 0 & e^x & x(x) \end{vmatrix}$$

$$W_1 = x(x) e^{3x}$$

$$W_2 = -\frac{2e^{2x}}{2e^{3x}}$$

$$W_3 = \frac{e^x}{2e^{3x}}$$

$$u_1 = \int \frac{W_1}{W} = \int \frac{x(x) e^{3x}}{2e^{3x}} = \frac{1}{2} \int \frac{e^{2x}}{1+e^x}$$

$$= \frac{1}{2} \left(\frac{1+e^x}{1+e^x} - \frac{1}{1+e^x} \right) e^x$$

$$= \frac{1}{2} \int e^x - \frac{1}{e^x} \int \frac{e^x}{1+e^x}$$

$$u_1 = \frac{e^x}{2} - \frac{\ln(1+e^x)}{2} + c_4$$

$$u_2 = \int \frac{W_2}{W} = \int \frac{2e^{2x}}{2e^{3x}} \cdot \frac{e^{2x}}{1+e^x} = -\int \frac{e^x}{1+e^x} = -\ln(1+e^x) + c_5$$

$$u_3 = \int \frac{W_3}{W} = \int \frac{e^x}{2e^{3x}} \frac{e^{2x}}{1+e^x} = \frac{1}{2} \int \frac{1}{1+e^x} = \frac{1}{2} \int \frac{1+e^x}{1+e^x} - \frac{e^x}{1+e^x}$$

$$= \frac{x}{2} - \frac{\ln(1+e^x)}{2} + c_6$$

$$y = c_1 + c_2 e^x + c_3 e^{2x} - \frac{\ln(1+e^x)}{2} + e^x \ln\left(\frac{\sqrt{e}}{1+e^x}\right) + \frac{e^{2x}}{2} \ln\left(\frac{e^x}{1+e^x}\right)$$